

Student Number	
Mark / 64	

Chemistry

**Final Examination
Preliminary Course • 2003**

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A Data Sheet and a Periodic Table are provided at the back of this paper
- Write your Student Number at the top of this page

Total Marks – 64

Part A – 10 marks

- Attempt Questions 1 – 10
- Allow about 10 minutes for this part

Part B – 54 marks

- Attempt Questions 11 – 23
- Allow about 110 minutes for this part

Part A – 10 marks

Attempt Questions 1–10

Allow about 10 minutes for this part

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct ↗

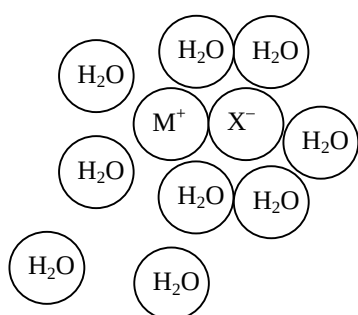
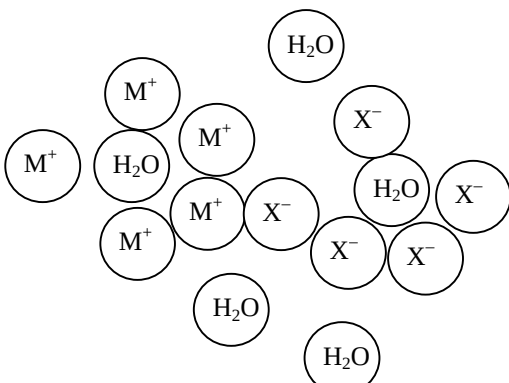
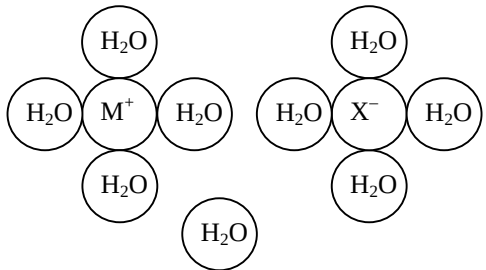
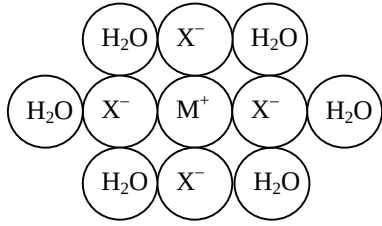
Answer Box for Questions 1–10				
1	A ☐	B ☐	C ☐	D ☐
2	A ☐	B ☐	C ☐	D ☐
3	A ☐	B ☐	C ☐	D ☐
4	A ☐	B ☐	C ☐	D ☐
5	A ☐	B ☐	C ☐	D ☐
6	A ☐	B ☐	C ☐	D ☐
7	A ☐	B ☐	C ☐	D ☐
8	A ☐	B ☐	C ☐	D ☐
9	A ☐	B ☐	C ☐	D ☐
10	A ☐	B ☐	C ☐	D ☐

► Mark your answers for Questions 1 – 10 in the Answer Box on page 1.

1 Which is a toxic gas pollutant from the incomplete combustion of petrol in cars?

- (A) ammonia
- (B) carbon monoxide
- (C) soot
- (D) carbon dioxide

2 Which diagram shows the complete dissolution of an ionic solid (M^+X^-) in water?

(A) 	(B) 
(C) 	(D) 

3 Which shows the correct percentage of water in the corresponding sphere?

	sphere	percentage water
(A)	atmosphere	0.5 – 10%
(B)	hydrosphere	90 – 94%
(C)	lithosphere	< 10%
(D)	biosphere (living matter)	45 – 90%

4 Which statement is true for a system undergoing an exothermic reaction?

- (A) The final energy content of the system is greater than the initial energy content.
- (B) The activation energy has a negative value.
- (C) The temperature decreases.
- (D) The ΔH has a negative value.

5 Organisms living in an aquatic habitat experience less temperature extremes than nearby organisms living on the land. Which factor explains the moderating effect of the water?

- (A) extensive hydrogen bonding
- (B) strong dispersion forces
- (C) high viscosity
- (D) high density

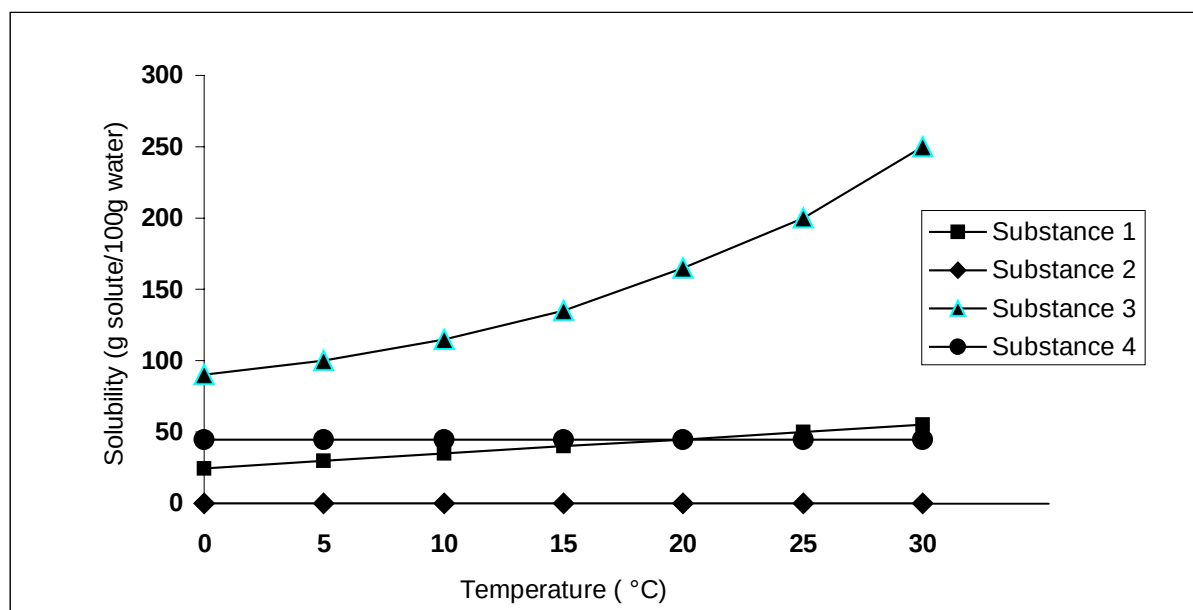
6 What is the mass of magnesium oxide (MgO) produced by burning 6.075 g of magnesium?

- (A) 0.250 g
- (B) 6.075 g
- (C) 10.075 g
- (D) 40.300 g

7 What is the mass of potassium hydroxide (KOH) needed to prepare 200 mL of a 0.25 mol L^{-1} solution?

- (A) 2.8 g
- (B) 28 g
- (C) 280 g
- (D) 2800 g

- 8 The graph shows the solubilities of four solid substances in water at different temperatures.



Which substance would be a covalent network solid?

- (A) Substance 1
(B) Substance 2
(C) Substance 3
(D) Substance 4
- 9 What is the number of molecules present in 22 g of CO_2 at 298 K and 100 kPa?
- (A) 3.0×10^{23}
(B) 6.0×10^{23}
(C) 12×10^{23}
(D) $6.0 \times 10^{11.5}$
- 10 A slight increase in temperature often causes a dramatic increase in the rate of a chemical reaction. Which statement best explains this effect?
- (A) The average frequency of collisions between particles increases.
(B) The ΔH for the reaction decreases.
(C) The activation energy is lowered.
(D) The number of molecules with energy greater than the activation energy increases.

Part B – 54 marks

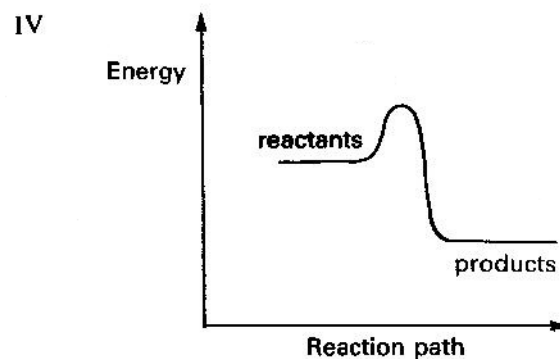
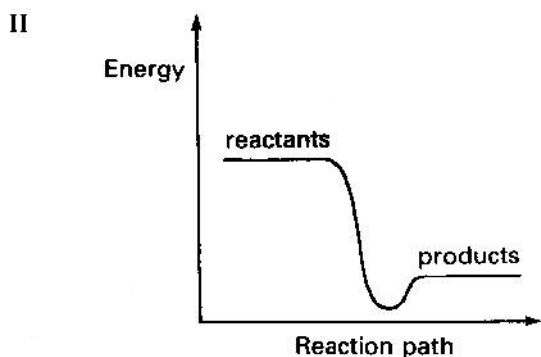
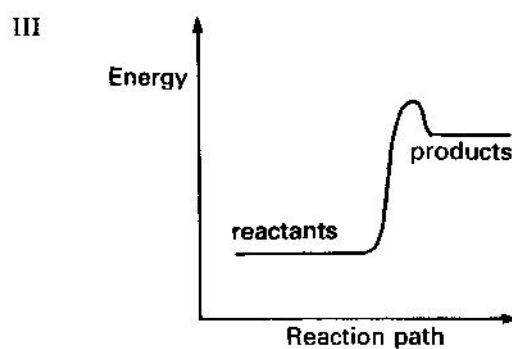
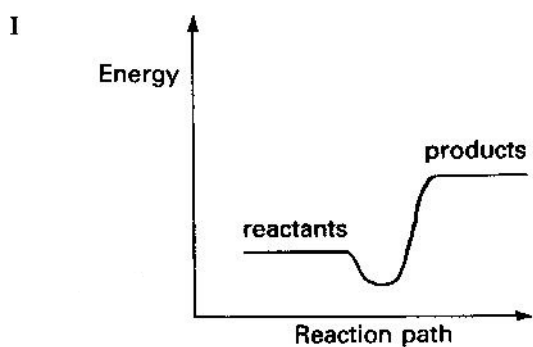
Attempt Questions 11 – 23

Allow about 110 minutes for this part

► **Show all relevant working in questions involving calculations.**

Question 11 (3 marks)

The graphs show the energy changes during the course of four different situations (I – IV)...



- (a) Which graph could correspond to the reaction: $\text{NO}_2(\text{g}) \rightarrow \frac{1}{2}\text{N}_2(\text{g}) + \text{O}_2(\text{g})$ $\Delta H = -33.7 \text{ kJ mol}^{-1}$?

_____ (1 mark)

- (b) Which graph could correspond to the melting of ice? _____ (1 mark)

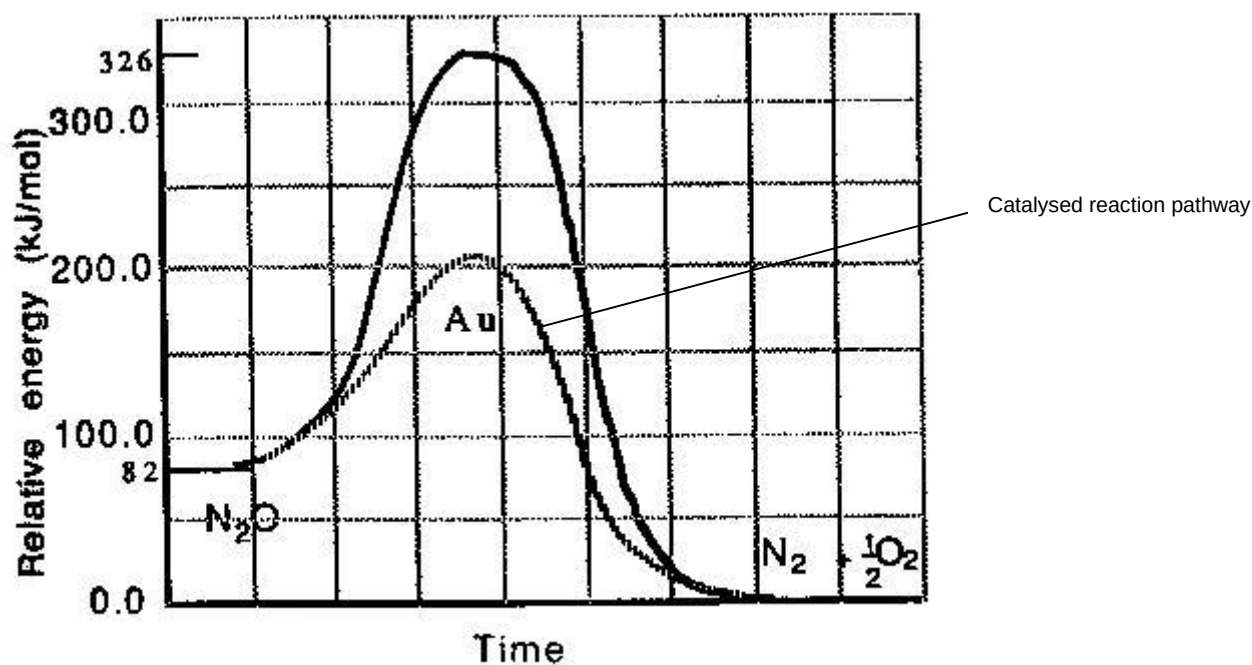
- (c) Which graph could correspond to the combustion of methane? _____ (1 mark)

Question 12 (4 marks)

- (a) Explain how fine coal dust in a coal mine can be an explosive hazard. **(2 marks)**

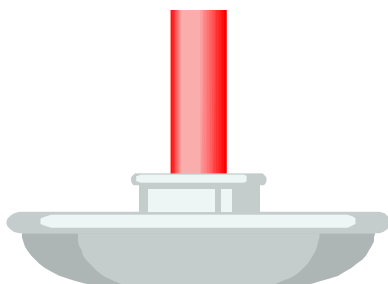
- (b) Suggest one safety feature adopted by industries to avoid dust explosions. **(1 mark)**

- (c) Dinitrogen monoxide can be thermally decomposed to nitrogen and oxygen.
The reaction is catalysed by gold... $\text{N}_2\text{O}_{(\text{g})} \rightarrow \text{N}_{2(\text{g})} + \frac{1}{2}\text{O}_{2(\text{g})} \quad \Delta H = -82 \text{ kJ mol}^{-1}$



What is the activation energy for the reverse reaction... $\text{N}_{2(\text{g})} + \frac{1}{2}\text{O}_{2(\text{g})} \rightarrow \text{N}_2\text{O}_{(\text{g})}$
in the absence of the gold catalyst? **(1 mark)**

Question 13 (3 marks)



A candle without a wick will not burn



A candle with a wick will burn readily

- (a) Identify two physical changes occurring during the burning of a candle. **(2 marks)**

- (b) Explain why only the candle with the wick burns. **(1 mark)**

Question 14 (6 marks)

- (a) Construct the Lewis electron dot structures for ammonia and water. **(2 marks)**

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Question 14 continues on page 8.

Question 14 (continued)

- (b) Describe the shape of the hydrogen sulfide molecule and explain why hydrogen sulfide has this shape. **(2 marks)**

- (c) Draw a diagram of an ammonia molecule showing its correct shape. **(1 mark)**
Identify the shape. **(1 mark)**



Question 15 (3 marks)

- (a) Which of hydrogen sulfide and water has the higher boiling point? **(1 mark)**

- (b) Explain your answer to (a). **(2 marks)**

Question 16 (3 marks)

- (a) What property of water enables an insect to walk on water? **(1 mark)**

- (b) Explain the nature of the property of water identified in (a) in terms of intermolecular forces. **(2 marks)**

Question 17 (3 marks)

You have already done an experiment where you used water's ability to absorb heat to measure energy changes in reactions.

- (a) Identify or describe the apparatus you used to perform the experiment. **(1 mark)**

- (b) Identify one measurement required to do this experiment. **(1 mark)**

- (c) Write the equation you used to determine the amount of heat absorbed. **(1 mark)**

Question 18 (5 marks)

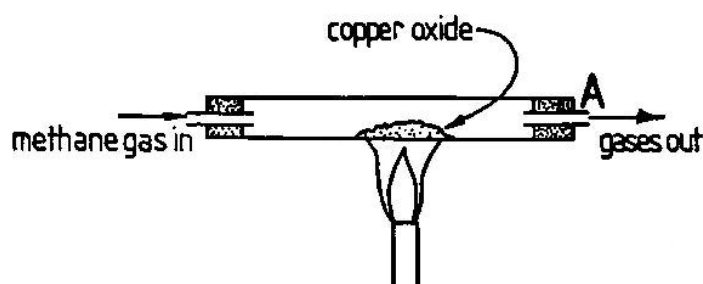
A cargo helicopter accidentally dropped 1200 kg of chemical in a pond containing 50,000 litres of water. When the chemical dissolved in the pond, the temperature increased.

- (a) How much heat was released to the water in the pond, if the water temperature in the pond increased from 15°C to 21°C? **(2 marks)**

- (b) Outline three implications (other than directly killing organisms) for aquatic life subjected to thermal pollution. **(3 marks)**

Question 19 (5 marks)

The diagram shows methane gas passing over heated copper(II) oxide reacting to produce copper metal and gaseous products of carbon dioxide and water vapour which leave the apparatus at A...



- (a) Write the balanced formulae equation for the reaction of copper(II) oxide with methane (CH_4). **(1 mark)**

- (b) If 15.9 g of copper(II) oxide is completely reacted, calculate the mass of copper metal formed. **(2 marks)**

- (c) Calculate the percentage of copper in the copper(II) oxide sample. **(1 mark)**

- (d) Calculate the volume of CO_2 produced at 25°C and 100 kPa from 15.9 g of copper(II) oxide. **(1 mark)**

Question 20 (4 marks)

- (a) Legislation states that the concentration of alcohol, $\text{C}_2\text{H}_5\text{OH}$, in the blood of an experienced driver of a motor car is not to exceed 0.05% (w/v).

Calculate the corresponding $\text{C}_2\text{H}_5\text{OH}$ concentration in the blood in moles per litre. **(2 marks)**

- (b) Identify a different measurement of concentration to that mentioned above (i.e. w/v) and describe a use for this measurement. **(2 marks)**

Question 21 (4 marks)

- (a) Calculate the mass of sodium sulfate required to prepare 50.0 mL of 0.150 mol L^{-1} solution. **(1 mark)**

- (b) What volume of this solution would you need to dilute, to prepare 125 mL of $0.0500 \text{ mol L}^{-1}$ solution? **(1 mark)**

- (c) What is the concentration of the sodium ions and sulfate ions in the $0.0500 \text{ mol L}^{-1}$ solution? **(2 marks)**

Question 22 (4 marks)

The table shows data for the compound hydrazine...

Composition	Hydrazine is a compound of nitrogen and hydrogen
Complete combustion of gaseous hydrazine at 400 K and 100 kPa	$\begin{array}{ccccccc} \text{hydrazine (g)} & + & \text{oxygen (g)} & \rightarrow & \text{nitrogen dioxide (g)} & + & \text{water (g)} \\ 1.0 \text{ L} & & 3.0 \text{ L} & & 2.0 \text{ L} & & 2.0 \text{ L} \end{array}$

- (a) Explain how the data for combustion illustrates Gay-Lussac's Law of Combining Gas Volumes. **(1 mark)**

- (b) Determine the molecular formula of hydrazine.
Show all working. **(2 marks)**

- (c) What is the empirical formula of hydrazine? **(1 mark)**

Question 23 (7 marks)

2.08 g of barium chloride was dissolved in water to make 50.0 mL of solution and then added to 50.0 mL of an aqueous solution containing 2.84 g of sodium sulfate. A white precipitate formed.

- (a) Write the net ionic equation for the reaction forming the precipitate. **(1 mark)**
► Use the solubility table below to determine the identity of the precipitate.

ANION	CATION	COMPOUND
All	Group I metals	soluble
All	Ammonium, NH_4^+	soluble
Nitrate, NO_3^-	All	soluble
Acetate/ethanoate CH_3COO^-	All except Ag^+	soluble
Chloride, Cl^- Bromide, Br^- Iodide, I^-	Ag^+ , Pb^{2+} , Hg_2^{2+} , Cu^+	insoluble
	All others	soluble
Sulfate, SO_4^{2-}	Ca^{2+} , Sr^{2+} , Ba^{2+} , Pb^{2+} , Ag^+ , Hg_2^{2+}	insoluble
	All others	soluble
Sulfide, S^{2-}	Group I and II metals, NH_4^+	soluble
	All others	insoluble
Hydroxide, OH^-	Group I metals, NH_4^+ , Sr^{2+} , Ba^{2+}	soluble
	All others	insoluble
Carbonate, CO_3^{2-} Phosphate, PO_4^{3-} Sulfite, SO_3^{2-}	Group I metals, NH_4^+	soluble
	All others	insoluble

Question 23 continues on page 15

Question 23 (continued)

(b) What is the mass of the precipitate formed? Show working. **(4 marks)**

(c) Calculate the concentration (mol L^{-1}) of sulfate ions in the final solution. Show working. **(2 marks)**

HIGHER SCHOOL CERTIFICATE EXAMINATION
Chemistry

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+] \qquad \Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS																KEY		Atomic Number		Atomic Weight		Symbol of element		Name of element		Atomic Number		Atomic Weight		Symbol of element		Name of element							
																79		Au		197.0		Gold																	
																5		B		10.81		Boron																	
																6		C		12.01		Carbon																	
																7		N		14.01		Nitrogen																	
																8		O		16.00		Oxygen																	
																9		F		19.00		Fluorine																	
																10		Ne		20.18		Neon																	
																11		Na		22.99		Sodium																	
																12		Mg		24.31		Magnesium																	
																13		Al		26.98		Aluminum																	
																14		Si		28.09		Silicon																	
																15		P		30.97		Phosphorus																	
																16		S		32.07		Sulfur																	
																17		Cl		35.45		Chlorine																	
																18		Ar		39.95		Argon																	
																19		K		39.10		Potassium																	
																20		Ca		40.08		Calcium																	
																21		Sc		44.96		Scandium																	
																22		Ti		47.87		Titanium																	
																23		V		50.94		Vanadium																	
																24		Cr		52.00		Chromium																	
																25		Mn		54.94		Manganese																	
																26		Fe		55.85		Iron																	
																27		Co		58.93		Cobalt																	
																28		Ni		58.69		Nickel																	
																29		Cu		63.55		Copper																	
																30		Zn		65.39		Zinc																	
																31		Ga		69.72		Gallium																	
																32		Ge		72.61		Germanium																	
																33		As		74.92		Arsenic																	
																34		Se		78.96		Selenium																	
																35		Br		79.90		Bromine																	
																36		Kr		83.80		Krypton																	
																37		Rb		85.47		Rubidium																	
																38		Sr		87.62		Strontium																	
																39		Y		88.91		Yttrium																	
																40		Zr		91.22		Zirconium																	
																41		Nb		92.91		Niobium																	
																42		Mo		95.94		Molybdenum																	
																43		Tc		[98.91]		Technetium																	
																44		Ru		101.1		Ruthenium																	
																45		Rh		102.9		Rhodium																	
																46		Pd		106.4		Palladium																	
																47		Ag		107.9		Silver																	
																48		Cd		112.4		Cadmium																	
																49		In		114.8		Indium																	
																50		Sn		118.7		Tin																	
																51		Sb		121.8		Antimony																	
																52		Te		127.6		Tellurium																	
																53		I		126.9		Iodine																	
																54		Xe		131.3		Xenon																	
																55		Cs		132.9		Caesium																	
																56		Ba		137.3		Barium																	
																57-71																							
																72		Hf		178.5		Hafnium																	
																73		Ta		180.9		Tantalum																	
																74		W		183.8		Tungsten																	
																75		Re		186.2		Rhenium																	
																76		Os		190.2		Osmium																	
																77		Ir		192.2		Iridium																	
																78		Pt		195.1		Platinum																	
																79		Au		197.0		Gold																	
																80		Hg		200.6		Mercury																	
																81		Tl		204.4		Thallium																	
																82		Pb		207.2		Lead																	
																83		Bi		209.0		Bismuth																	
																84		Po		[210.0]		Polonium																	
																85		At		[210.0]		Astatine																	
																86		Rn		[222.0]		Radon																	
																87		Fr		[223.0]		Francium																	
																88		Ra		[226.0]		Radium																	
																89-103																							
																104		Rf		[261.1]		Rutherfordium																	
																105		Db		[262.1]		Dubnium																	
																106		Sg		[263.1]		Seaborgium																	
																107		Bh		[264.1]		Bohrium																	
																108		Hs		[265.1]		Hassium																	
																109		Mt		[268]		Meitnerium																	
																110		Un		—		Ununium																	
																111		Uu		—		Ununium																	
																112		Uub		—		Unbium																	
																113		Uut		—		Untrium																	
																114		Uuq		—		Unquadium																	
																115		Uuh		—		Unhexium																	
																116		Uuh		—		Unhexium																	
																117		Uus		—		Unseptium																	
																118		Uuo		—		Unoctium																	

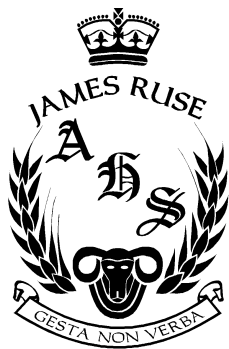
Lanthanides

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [146.9] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
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Actinides

89 Ac [227.0] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237.0] Neptunium	94 Pu [239.1] Plutonium	95 Am [241.1] Americium	96 Cm [244.1] Curium	97 Bk [249.1] Berkelium	98 Cf [252.1] Californium	99 Es [257.1] Einsteinium	100 Fm [257.1] Fermium	101 Md [258.1] Mendelevium	102 No [259.1] Nobelium	103 Lr [262.1] Lawrencium
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Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.



ANSWERS and MARKING SCHEME

Chemistry

**Final Examination
Preliminary Course • 2003**

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Student Number at the top of this page

Total Marks – 64

Part A – 10 marks

- Attempt Questions 1 – 10
- Allow about 10 minutes for this part

Part B – 54 marks

- Attempt Questions 11 – 23
- Allow about 110 minutes for this part

Part A – 10 marks

Attempt Questions 1–10

Allow about 10 minutes for this part

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct ↖

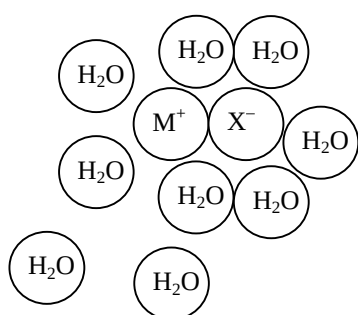
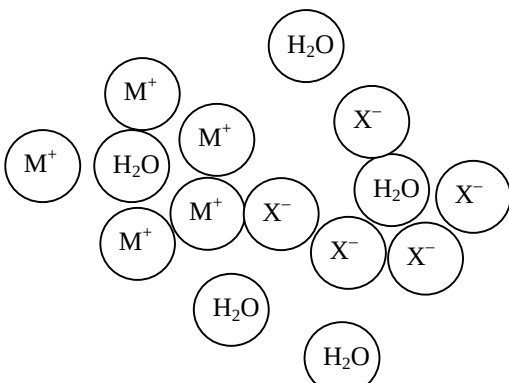
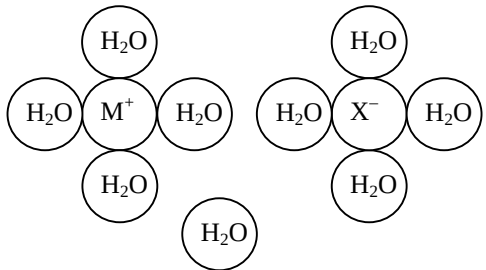
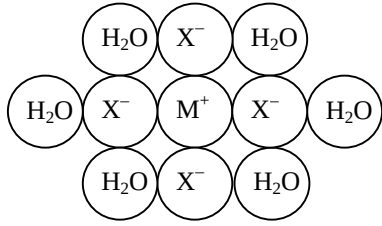
Answer Box for Questions 1 – 10				
1	A ☐	B ☒	C ☐	D ☐
2	A ☐	B ☐	C ☒	D ☐
3	A ☐	B ☐	C ☒	D ☐
4	A ☐	B ☐	C ☐	D ☒
5	A ☒	B ☐	C ☐	D ☐
6	A ☐	B ☐	C ☒	D ☐
7	A ☒	B ☐	C ☐	D ☐
8	A ☐	B ☒	C ☐	D ☐
9	A ☒	B ☐	C ☐	D ☐
10	A ☐	B ☐	C ☐	D ☒

► Mark your answers for Questions 1 – 10 in the Answer Box on page 1.

1 Which is a toxic gas pollutant from the incomplete combustion of petrol in cars?

- (A) ammonia
- (B) carbon monoxide
- (C) soot
- (D) carbon dioxide

2 Which diagram shows the complete dissolution of an ionic solid (M^+X^-) in water?

(A) 	(B) 
(C) 	(D) 

3 Which shows the correct percentage of water in the corresponding sphere?

	sphere	percentage water
(A)	atmosphere	0.5 – 10%
(B)	hydrosphere	90 – 94%
(C)	lithosphere	< 10%
(D)	biosphere (living matter)	45 – 90%

4 Which statement is true for a system undergoing an exothermic reaction?

- (A) The final energy content of the system is greater than the initial energy content.
- (B) The activation energy has a negative value.
- (C) The temperature decreases.
- (D) The ΔH has a negative value.

5 Organisms living in an aquatic habitat experience less temperature extremes than nearby organisms living on the land. Which factor explains the moderating effect of the water?

- (A) extensive hydrogen bonding
- (B) strong dispersion forces
- (C) high viscosity
- (D) high density

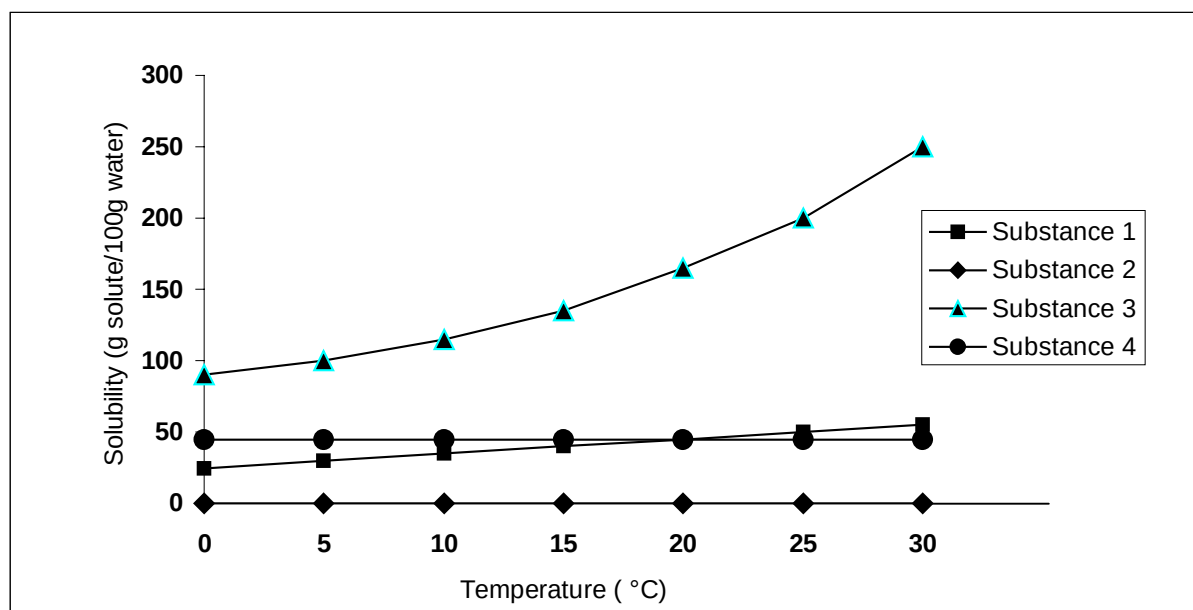
6 What is the mass of magnesium oxide (MgO) produced by burning 6.075 g of magnesium?

- (A) 0.250 g
- (B) 6.075 g
- (C) 10.075 g
- (D) 40.300 g

7 What is the mass of potassium hydroxide (KOH) needed to prepare 200 mL of a 0.25 mol L^{-1} solution?

- (A) 2.8 g
- (B) 28 g
- (C) 280 g
- (D) 2800 g

- 8 The graph shows the solubilities of four solid substances in water at different temperatures.



Which substance would be a covalent network solid?

- (A) Substance 1
(B) Substance 2
(C) Substance 3
(D) Substance 4
- 9 What is the number of molecules present in 22 g of CO_2 at 298 K and 100 kPa?
- (A) 3.0×10^{23}
(B) 6.0×10^{23}
(C) 12×10^{23}
(D) $6.0 \times 10^{11.5}$
- 10 A slight increase in temperature often causes a dramatic increase in the rate of a chemical reaction. Which statement best explains this effect?
- (A) The average frequency of collisions between particles increases.
(B) The ΔH for the reaction decreases.
(C) The activation energy is lowered.
(D) The number of molecules with energy greater than the activation energy increases.

Part B – 54 marks

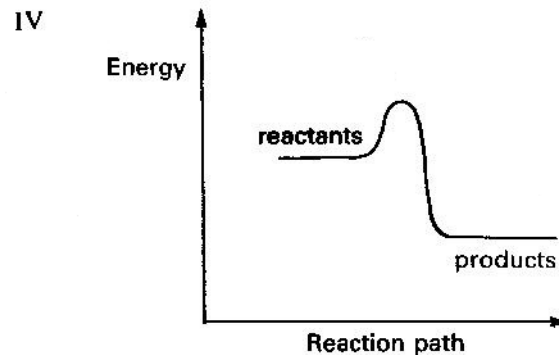
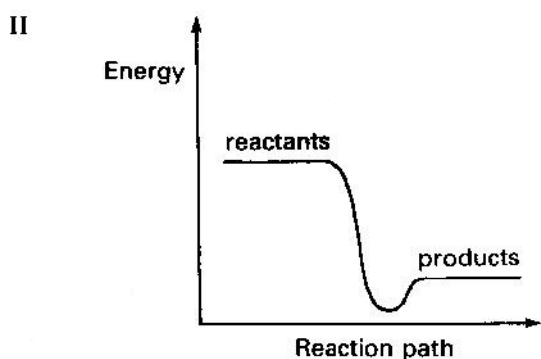
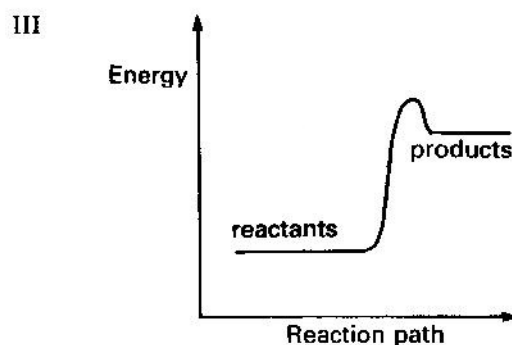
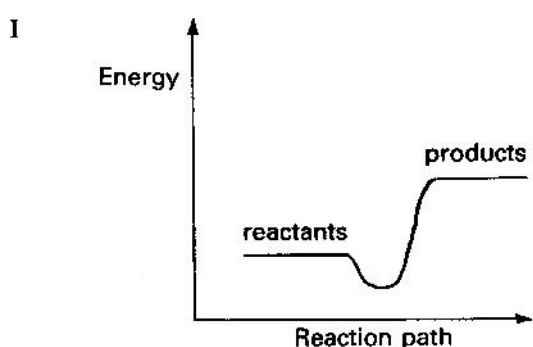
Attempt Questions 11 – 23

Allow about 110 minutes for this part

► **Show all relevant working in questions involving calculations.**

Question 11 O/C – P7 (3 marks)

The graphs show the energy changes during the course of four hypothetical reactions or processes (I – IV)...



- (a) Which graph could correspond to the reaction: $\text{NO}_2(\text{g}) \rightarrow \frac{1}{2}\text{N}_2(\text{g}) + \text{O}_2(\text{g})$ $\Delta H = -33.7 \text{ kJ mol}^{-1}$?

Graph IV (1 mark)

- (b) Which graph could correspond to the melting of ice? Graph III (1 mark)

- (c) Which graph could correspond to the combustion of methane? Graph IV (1 mark)

Question 12 (4 marks)

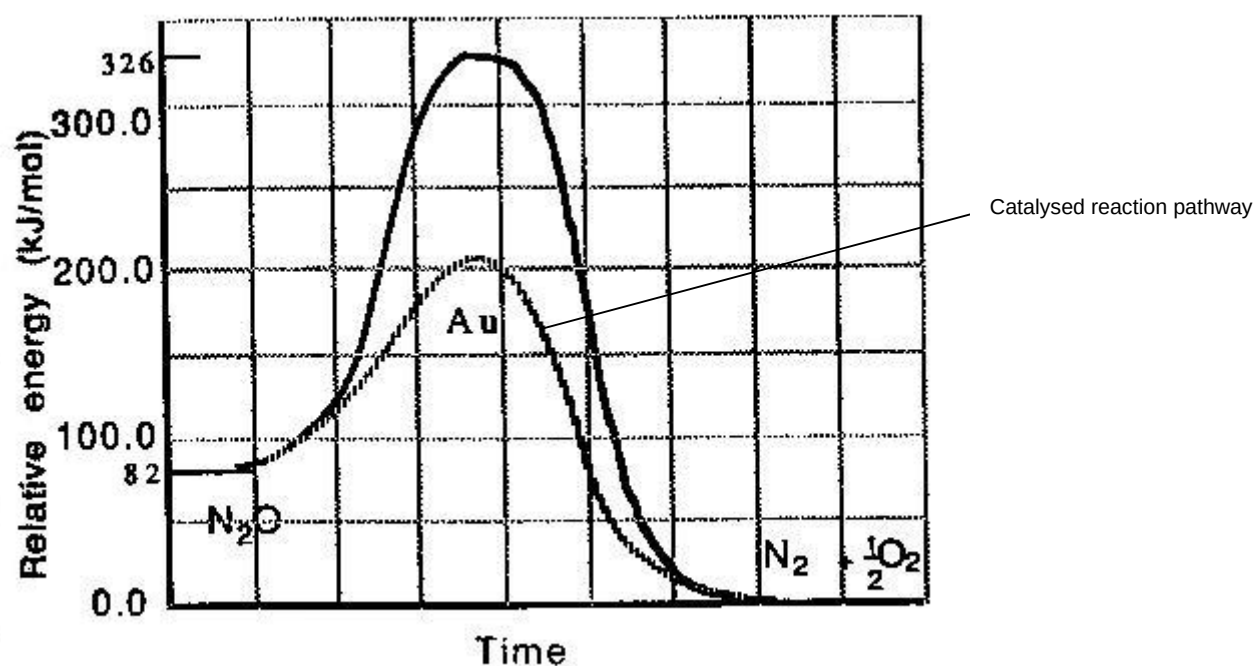
- (a) Explain how fine coal dust in a coal mine can be an explosive hazard. **O/C – P4 (2 marks)**

The total surface area of the particles is large (1 mark) and each particle has a ready supply of oxygen available. As the reaction is exothermic (1 mark), heat released supplies the activation energy to hasten further reactions and an explosion ensues.

- (b) Suggest one safety feature adopted by industries to avoid dust explosions. **O/C – P4 (1 mark)**

Ensuring there is no build-up of fine flammable particles by well-ventilating the worksite.

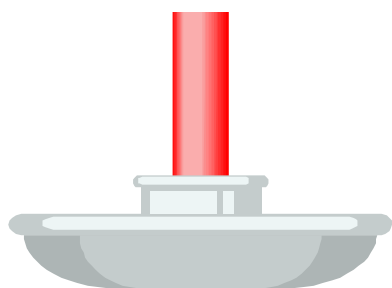
- (c) Dinitrogen monoxide can be thermally decomposed to nitrogen and oxygen.
The reaction is catalysed by gold... $\text{N}_2\text{O}_{(g)} \rightarrow \text{N}_{2(g)} + \frac{1}{2}\text{O}_{2(g)} \quad \Delta H = -82 \text{ kJ mol}^{-1}$



What is the activation energy for the reverse reaction... $\text{N}_{2(g)} + \frac{1}{2}\text{O}_{2(g)} \rightarrow \text{N}_2\text{O}_{(g)}$
in the absence of the gold catalyst? **O/C – P7, 14 (1 mark)**

326 kJ

Question 13 O/C – P7 (3 marks)



A candle without a wick will not burn



A candle with a wick will burn readily

- (a) Identify two physical changes occurring during the burning of a candle. **(2 marks)**

Melting or fusion or solid → liquid

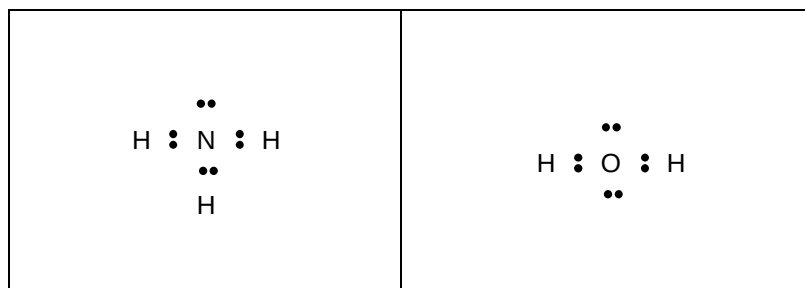
Vaporisation or liquid → gas

- (b) Explain why only the candle with the wick burns. **(1 mark)**

Wax cannot burn directly as a solid. A burning wick provides heat to melt and vapourise the wax which can then be fed up the wick (by capillary action) to sustain the flame and hence continue the combustion process.

Question 14 (6 marks)

- (a) Construct the Lewis electron dot structures for ammonia and water. **O/C – P7 (2 marks)**



Question 14 (continued)

- (b) Describe the shape of the hydrogen sulfide molecule and explain why hydrogen sulfide has this shape. **O/C – P14 (2 marks)**

Hydrogen sulfide is a bent molecule (1 mark) due to the central sulfur atom having two lone (non-bonding) pairs of electrons. This causes repulsion of the covalent bonds (1 mark).

► *The four pairs of electrons around the central sulfur atom will automatically orientate towards maximum repulsion resulting in a tetrahedral placement in 3D space. However, the shape of a molecule is determined by the placement of the bonded atoms, ∴ in this case, bent.*

- (c) Draw a diagram of an ammonia molecule showing its correct shape. **(1 mark) O/C – P14**
Identify the shape. **(1 mark)**

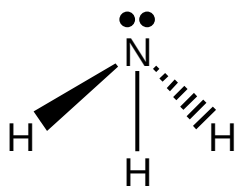


Diagram (1 mark)

Ammonia is a trigonal pyramid (1 mark)

► 'pyramidal' accepted as per Smith text

Question 15 (3 marks)

- (a) Which of hydrogen sulfide and water has the higher boiling point? **O/C – P14 (1 mark)**

Water has the higher boiling point.

- (b) Explain your answer to (a). **O/C – P14 (2 marks)**

Water forms two hydrogen bonds per molecule.

Hydrogen bonding is the strongest intermolecular force. (1 mark)

Hydrogen sulfide, however, has dipole–dipole forces (and dispersion forces).

These are not as strong as hydrogen bonding. (1 mark)

Question 16 (3 marks)

- (a) What property of water enables an insect to walk on water? **O/C – P14 (1 mark)**

Surface tension

- (b) Explain the nature of the property of water identified in (a) in terms of intermolecular forces.
O/C – P7 (2 marks)

Water molecules at the surface form hydrogen bonding only with water molecules that are adjacent and below. There are no molecules (or attractions) from above. (1 mark)

These unbalanced forces on surface molecules result in a downward or inward force acting on surface molecules, ∴ surface tension. (1 mark)

Question 17 O/C – P4, 7 (3 marks)

You have already done an experiment where you used water's ability to absorb heat to measure energy changes in reactions.

- (a) Identify or describe the apparatus you used to perform the experiment. **(1 mark)**

Calorimeter or plastic foam cup + thermometer

- (b) Identify one measurement required to do this experiment. **(1 mark)**

- **Mass/volume of water/solution in the calorimeter.**
 - **Change in temperature of the water/solution in the calorimeter.**
 - **Mass of reagent involved in the reaction.**
- **Any one of the above.**

- (c) Write the equation you used to determine the amount of heat absorbed. **(1 mark)**

- **$q = m C \Delta T$**
 - **$\Delta H = - m C \Delta T$**
 - **$\Delta H = m C \Delta T$**
- **Any one of the above.**

Question 18 O/C – P4, 7, 10, 16 (5 marks)

A cargo helicopter accidentally dropped 1200 kg of chemical in a pond containing 50,000 litres of water. When the chemical dissolved in the pond, the temperature increased.

- (a) How much heat was released to the water in the pond, if the water temperature in the pond increased from 15°C to 21°C? **(2 marks)**

$$q = m C \Delta T = 50,000 \text{ kg} \times 4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1} \times (21^\circ\text{C} - 15^\circ\text{C}) = 1.254 \times 10^9 \text{ J} = \underline{1.3 \times 10^6 \text{ kJ}}$$

Consistent use of units = 1 mark

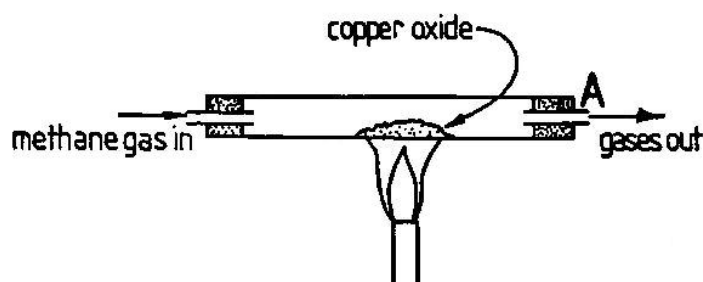
Correct calculation = 1 mark

- (b) Outline three implications (other than directly killing organisms) for aquatic life subjected to thermal pollution. **(3 marks)**

- Less dissolved oxygen causing stress to organism
- Increased metabolic rates which increases the demand for oxygen and so aggravates the low dissolved oxygen problem
- Fish eggs do not develop or hatch if temperature is too high
- False temperature cues are given to aquatic life,
∴ setting of migration and spawning at the wrong times of the year.

Question 19 O/C – P10, 13, 14 (5 marks)

The diagram shows methane gas passing over heated copper(II) oxide reacting to produce copper metal and gaseous products of carbon dioxide & water vapour which leave the apparatus at A...



- (a) Write the balanced formulae equation for the reaction of copper(II) oxide with methane (CH₄). **(1 mark)**



- (b) If 15.9 g of copper(II) oxide is reacted completely, calculate the mass of copper metal formed. **(2 marks)**

$$\text{moles CuO} = n = m \div M = 15.9 \text{ g} \div 79.55 \text{ g mol}^{-1} = \underline{0.200 \text{ mole}} \quad (1 \text{ mark})$$

$$\text{moles CuO} = \text{moles Cu}$$

$$\text{mass Cu} = m = n \times M = 0.200 \text{ mol} \times 63.55 \text{ g mol}^{-1} = \underline{12.7 \text{ g}} \quad (1 \text{ mark})$$

- (c) Calculate the percentage of copper in the copper(II) oxide sample. **(1 mark)**

$$\text{Cu}\% = \text{mass Cu} \div \text{mass CuO} \times 100 = 12.7 \text{ g} \div 15.9 \text{ g} \times 100 = \underline{79.9\%}$$

- (d) Calculate the volume of CO₂ produced at 25 °C and 100 kPa from 15.9 g of copper(II) oxide. **(1 mark)**

$$\text{moles CO}_2 = \frac{1}{2} \text{ moles CuO} = \frac{1}{2} \times 0.200 \text{ mol} = \underline{0.100 \text{ mol}}$$

$$\text{Volume CO}_2 = \text{moles} \times \text{molar volume} = 0.100 \text{ mol} \times 24.79 \text{ L mol}^{-1} = \underline{2.48 \text{ L}}$$

Question 20 O/C – P13 (4 marks)

- (a) Legislation states that the concentration of alcohol, $\text{C}_2\text{H}_5\text{OH}$, in the blood of an experienced driver of a motor car is not to exceed 0.05% (w/v).

Calculate the corresponding $\text{C}_2\text{H}_5\text{OH}$ concentration in the blood in moles per litre. **(2 marks)**

$$0.05\% \text{ (w/v)} = 0.05 \text{ g/100 mL} \quad (1 \text{ mark})$$

$$\text{moles ethanol} = 0.05 \text{ g} \div 46.068 \text{ g/mol} = 1.085 \times 10^{-3} \text{ mol}$$

$$\therefore [\text{C}_2\text{H}_5\text{OH}] = 1.085 \times 10^{-3} \text{ mol} \div 0.100 \text{ L} = \underline{0.01 \text{ mol L}^{-1}} \quad (1 \text{ mark})$$

- (b) Identify a different measurement of concentration to those mentioned above (i.e. w/v) and describe a use for this measurement. **(2 marks)**

ppm (or ppb, v/v, w/w) (1 mark)

ppm is a convenient unit for measuring concentrations of pollutants, e.g. lead ions in waterways. (1 mark)

Question 21 O/C – P10, 14 (4 marks)

- (a) Calculate the mass of sodium sulfate required to prepare 50.0 mL of 0.150 mol L^{-1} solution. **(1 mark)**

$$\text{mol of Na}_2\text{SO}_4 = 0.15 \text{ mol L}^{-1} \times 50 \times 10^{-3} \text{ L} = \underline{0.0075 \text{ mol}}$$

$$\text{mass of Na}_2\text{SO}_4 = 0.0075 \text{ mol} \times 142.05 \text{ g/mol} = 1.065 = \underline{1.07 \text{ g}}$$

- (b) What volume of this solution would you need to dilute, to prepare 125 mL of $0.0500 \text{ mol L}^{-1}$ solution? **(1 mark)**

$$c_1V_1 = c_2V_2; V_1 = c_2V_2 \div c_1 = 0.0500 \text{ mol L}^{-1} \times 0.125 \text{ L} \div 0.150 \text{ mol L}^{-1} = 0.0417 \text{ L} = \underline{41.7 \text{ mL}}$$

- (c) What is the concentration of the sodium ions and sulfate ions in the $0.0500 \text{ mol L}^{-1}$ solution? **(2 marks)**



$$\therefore [\text{Na}^+] = 0.0500 \text{ mol L}^{-1} \times 2 = \underline{0.100 \text{ mol L}^{-1}} \quad (1 \text{ mark})$$

$$\therefore [\text{SO}_4^{2-}] = \underline{0.0500 \text{ mol L}^{-1}} \quad (1 \text{ mark})$$

Question 22 O/C – P1, 2, 13 (4 marks)

The table shows data for the compound hydrazine...

Composition	Hydrazine is a compound of nitrogen and hydrogen
Complete combustion of gaseous hydrazine at 400 K and 100 kPa	$\begin{array}{ccccccc} \text{hydrazine (g)} & + & \text{oxygen (g)} & \rightarrow & \text{nitrogen dioxide (g)} & + & \text{water (g)} \\ 1.0 \text{ L} & & 3.0 \text{ L} & & 2.0 \text{ L} & & 2.0 \text{ L} \end{array}$

- (a) Explain how the data for combustion illustrates Gay-Lussac's Law of Combining Gas Volumes. **(1 mark)**

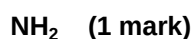
The gas volumes are in simple whole number ratios.

- (b) Determine the molecular formula of hydrazine.
Show all working. **(2 marks)**



$$\begin{aligned} \therefore x &= 2; y = 4 \\ \therefore \text{N}_2\text{H}_4 & \quad \textbf{(1 mark)} \end{aligned}$$

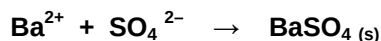
- (c) What is the empirical formula of hydrazine? **(1 mark)**



Question 23 O/C – P10, 13, 14 (7 marks)

2.08 g of barium chloride was dissolved in water to make 50.0 mL of solution and then added to 50.0 mL of an aqueous solution containing 2.84 g of sodium sulfate. A white precipitate formed.

- (a) Write the net ionic equation for the reaction forming the precipitate. **(1 mark)**



- (b) What is the mass of the precipitate formed? Show working. **(4 marks)**

$$\text{moles Ba}^{2+} = \text{moles BaCl}_2 = 2.08 \text{ g} \div 208.2 \text{ g/mol} = \underline{0.00999 \text{ or } 0.0100 \text{ mol}} \quad (1 \text{ mark})$$

$$\text{moles SO}_4^{2-} = \text{moles of Na}_2\text{SO}_4 = 2.84 \text{ g} \div 142.05 \text{ g/mol} = \underline{0.0200 \text{ mol}} \quad (1 \text{ mark})$$

$$\therefore \text{moles BaSO}_4 \text{ ppt.} = \underline{0.0100 \text{ mol}} \quad (1 \text{ mark})$$

$$\text{mass BaSO}_4 = 0.0100 \text{ mol} \times 233.37 \text{ g/mol} = \underline{2.33 \text{ g}} \quad (1 \text{ mark})$$

- (c) Calculate the concentration (mol L^{-1}) of sulfate ions in the final solution. Show working. **(2 marks)**

$$\text{moles SO}_4^{2-} \text{ excess remaining in solution} = 0.0100 \text{ mol} \quad (1 \text{ mark})$$

$$\therefore [\text{SO}_4^{2-}] = 0.0100 \text{ mol} \div 0.100 \text{ L} = \underline{0.100 \text{ mol L}^{-1}} \quad (1 \text{ mark})$$